Roll No. | | | | | | | |

ODD END SEMESTER EXAMINATION DECEMBER – 2024

Name of the course: B. Tech
Name of Paper: Material science
Time: 3 Hours

Semester: 3rd
Paper code: TME302
Maximum Marks: 100

Note: -

(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks

Q1	(20marks)	
(a)	Explain the close packing in solids with a focus on FCC and HCP structures	CO1, CO2
(b)	Classify materials based on their structure and properties with suitable examples.	
(c)	Draw a (110), (211) plane indices and [121], [211] direction indices inside a cubic unit cell.	
Q2	(20 marks)	
(a)	Discuss the importance of phase diagrams in material selection and design.	
(b)	Differentiate between – (a) Edge Dislocation and Screw dislocation (b) Grain boundary and twin boundary (c) Frenkel and Schottky defect (d) Porosity and blow holes	CO2, CO3
(c)	Illustrate and describe the Pb-Sn and Al-Si phase diagrams in detail.	
Q3	(20 marks)	
(a)	A BCC crystal is used to measure the wavelength od some x -rays. The Bragg's angle for reflection from (111) plane is 23.2^0. What is the wavelength? The lattice parameter of the crystal is 3.25 A^0.	CO3, CO4
(b)	Draw the Iron- carbon phase diagram in details. Also, explain the invariants reactions.	
(c)	Explain the concept of critical radius in homogeneous nucleation and derive the expression for its determination, considering the balance of volumetric free energy change and surface energy.	
Q4	(20 marks)	
(a)	Draw and discuss time temperature transformation diagram and their applications.	C04, CO5
(b)	Describe the following terms- (a) Annealing (b) Normalizing (c) Quenching (d) Martempering (e) Austempering	
(c)	Explain the following- (a) Carburizing (b) Nitriding (c) Carbonitriding (d) Induction hardening (e) Flame hardening	
Q5	(20 marks)	
(a)	Describe toughness, resilience, and proof resilience using the stress-strain diagram of cast iron, highlighting their physical significance with appropriate examples	C05, CO6

(b)	Explain the following- (a) paramagnetic, diamagnetism (b) ferro and ferry-magnetism (c) spontaneous and stimulated emission.	
(c)	Discuss the Hall effect in electronic materials. Derive the expression for Hall voltage and explain how it can be used to determine the type of charge carriers and their concentration in a material	